# Hawai‘i – Coastal flooding ESF-6 playbook

**Data as of 2026-10-27**

**County:** Hawai‘i · **Hazard:** Coastal flooding

**Planning aid. Supports, does not replace, county emergency plans. Not an official shelter list.**

## 1. Summary

| | | |
|---|---|---|
| People at risk<br>**53,969** | Shelter demand (central)<br>**2,698** | Shelter demand low – high<br>**1,079 – 5,937** |
| Limited English (C)<br>**1,841** | Power-dependent (M)<br>**410** | Disability (I)<br>**7,255** |
| Age 65+ (I)<br>**12,718** | People without a vehicle (T)<br>**4,088** | Known 2026 sites usable<br>**10** |

Shelter-seeking rates: low 2%, central 5%, high 11.0%. CMIST counts are people at risk, from census-tract estimates; one person can appear in more than one count.

## 2. Scope: county total vs. one community

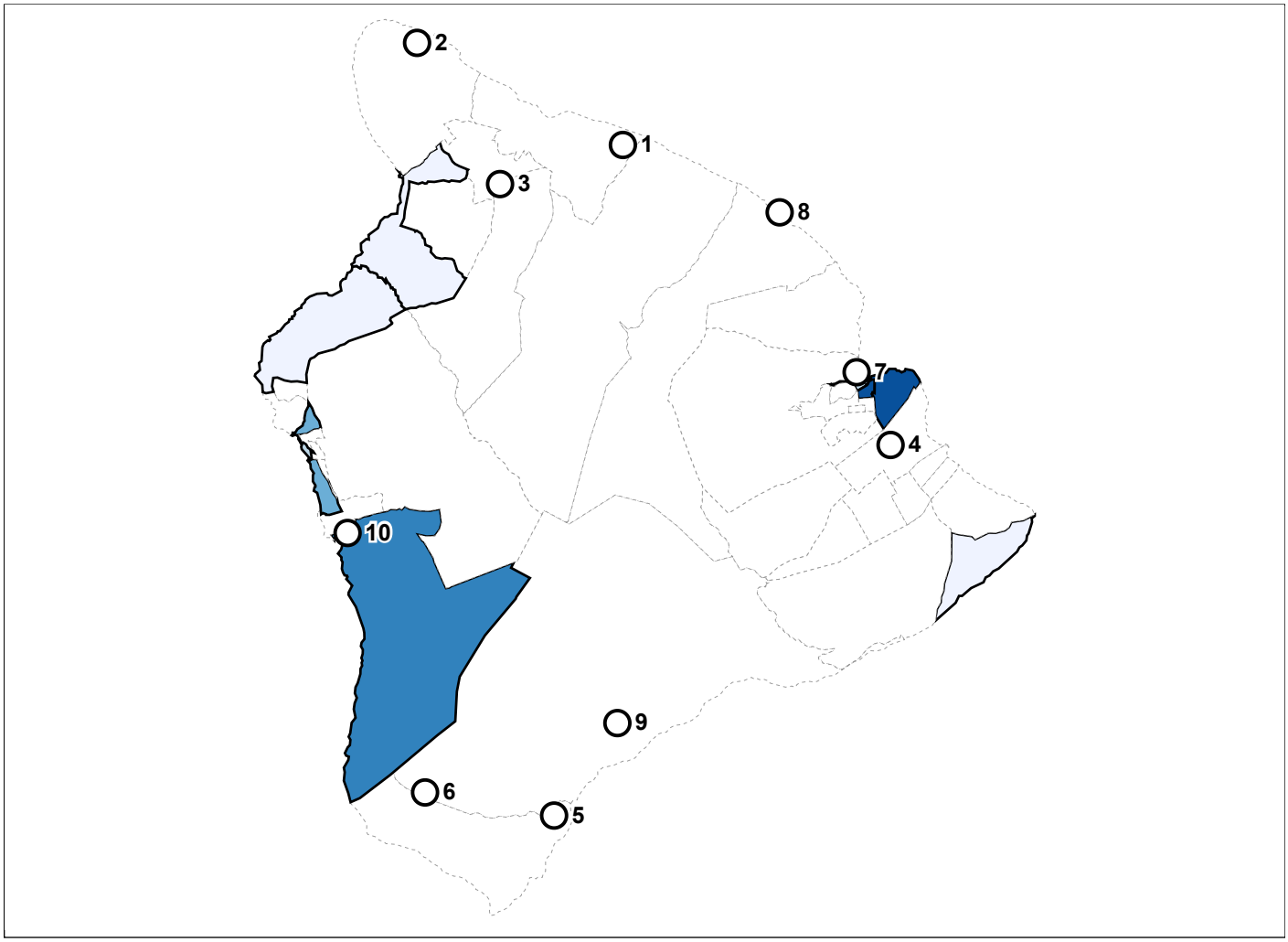
**County total = upper bound.** It assumes every at-risk tract is hit at once (top 25% of each county's tracts by NRI score). Real events usually hit fewer communities.

### Largest single community (highest-demand census tracts)

| Tract | At risk | Shelter low / central / high | Limited English | Power-dep. | Disability | 65+ | No vehicle |
|---|---|---|---|---|---|---|---|
| 15001021300 | 7,384 | 148 / **369** / 812 | 176 | 66 | 777 | 2,042 | 170 |
| 15001020600 | 6,802 | 136 / **340** / 748 | 67 | 45 | 1,261 | 1,394 | 387 |
| 15001021513 | 5,579 | 112 / **279** / 614 | 235 | 38 | 568 | 1,187 | 134 |

Plan a realistic one-community event with the first row.

## 3. Map: AFN need and known sites




- AFN need 0–975 people (class 1 of 5)
- AFN need 975–1,043 people (class 2 of 5)
- AFN need 1,043–1,173 people (class 3 of 5)
- AFN need 1,173–1,725 people (class 4 of 5)
- AFN need 1,725–2,219 people (class 5 of 5)
- Solid outline: tract in hazard footprint (at risk)
- Dashed outline: not in footprint
- Usable
- Check: flood (SFHA)
- Check: flood (tract-level)
- Flood risk undetermined (zone D)
- Excluded: tsunami zone

AFN need = power-dependent + disability + limited English + no-vehicle people at risk (people may be counted more than once). Numbers on markers match the sites table. No basemap: tract outlines only.

## 4. Known sites (opened in 2026 events)

| # | Status | Site | Address | Reason | Times opened 2026 |
|---|---|---|---|---|---|
| 1 | ✓ **Usable** | Honokaʻa Gym | 45-541 Lehua St., Honokaʻa | FEMA zone X: outside Special Flood Hazard Area | 1 |
| 2 | ✓ **Usable** | Ikuo Hisaoka Gym | 54-382 Kamehameha Park Road, Kapaʻau | FEMA zone X: outside Special Flood Hazard Area | 1 |
| 3 | ✓ **Usable** | Kahilu Town Hall | 67-1182 Lindsey Road, Waimea | FEMA zone X: outside Special Flood Hazard Area | 1 |
| 4 | ✓ **Usable** | Keaʻau Armory | 16-512 Volcano Road, Keaʻau | FEMA zone X: outside Special Flood Hazard Area | 1 |
| 5 | ✓ **Usable** | Nāʻālehu Elementary School Gym | 95-5545 Mamalahoa Highway, Nāʻālehu | FEMA zone X: outside Special Flood Hazard Area | 1 |
| 6 | ✓ **Usable** | Ocean View Community Center | 92-8924 Leilani Circle, Ocean View | FEMA zone X: outside Special Flood Hazard Area | 1 |
| 7 | ✓ **Usable** | Puʻuʻeo Community Center | 145 Wainaku St., Hilo | FEMA zone X: outside Special Flood Hazard Area | 1 |
| 8 | ✓ **Usable** | Pāpaʻaloa Park Annex | 35-1994 Government Main Road, Pāpaʻaloa | FEMA zone X: outside Special Flood Hazard Area | 1 |
| 9 | ✓ **Usable** | Robert N. Herkes Gym | 96-1219 Kamani St., Pāhala | FEMA zone X: outside Special Flood Hazard Area | 1 |
| 10 | ✓ **Usable** | Yano Hall | 82-6156 Māmalahoa Highway, Captain Cook | FEMA zone X: outside Special Flood Hazard Area | 1 |

Candidate public schools in this county usable for this hazard: **54**. Sites are places a county has opened before; not every site opens in every event. Always follow current county announcements.

## 5. Needs and proposed partners

Central estimate. Roles are **proposed – to be confirmed with partner**. Pledged capacity stays blank until a partner confirms it.

| Need | Quantity | Unit | Proposed lead | Proposed support | Pledged | Gap |
|---|---|---|---|---|---|---|
| Cots / sleeping space | 2,699 | cots | county_em | dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies | — | confirm with partner |
| Floor space (≤72 h) | 53,970 | sq ft | county_em | dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies | — | confirm with partner |
| Floor space (≤2 weeks) | 107,939 | sq ft | county_em | dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies | — | confirm with partner |
| Water | 2,699 | gal/day | red_cross | dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community | — | confirm with partner |
| Meals | 5,397 | meals/day | red_cross | dhs; county_em; doe; salvation_army; hawaii_voad; faith_based; business_community | — | confirm with partner |
| Toilets (≤72 h) | 68 | toilets | county_em | dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies | — | confirm with partner |
| Toilets (≤2 weeks) | 135 | toilets | county_em | dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies | — | confirm with partner |
| Shelter staff | 177 | staff per shift | county_em | dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; hawaii_voad; oha; salvation_army; philanthropy; business_community | — | confirm with partner |
| Power outlets for power-dependent residents | 21 | outlets | doh | county_em; health_care; utilities | — | confirm with partner |
| Accessible transport | 4,088 | people in no-vehicle households at risk | county_em | dhs; hiema; dot; business_community; other_agencies | — | confirm with partner |
| Pet space – dog households | 449 | households with a dog | county_em | hawaii_voad | — | confirm with partner |
| Pet space – cat households | 262 | households with a cat | county_em | hawaii_voad | — | confirm with partner |
| Children's area | 496 | children in shelter | county_em | dhs; hiema; doe; hspls; red_cross; faith_based; other_agencies; doh; health_care; philanthropy | — | confirm with partner |
| Interpreters – Ilocano | 23 | people with limited English in shelter | dhs | county_em; hiema; hspls; doh; oha; faith_based; other_agencies | — | confirm with partner |
| Interpreters – Other English-based Creole languages | 14 | people with limited English in shelter | dhs | county_em; hiema; hspls; doh; oha; faith_based; other_agencies | — | confirm with partner |
| Interpreters – Japanese | 10 | people with limited English in shelter | dhs | county_em; hiema; hspls; doh; oha; faith_based; other_agencies | — | confirm with partner |
| Interpreters – Spanish | 8 | people with limited English in shelter | dhs | county_em; hiema; hspls; doh; oha; faith_based; other_agencies | — | confirm with partner |
| Interpreters – Tagalog, Filipino | 6 | people with limited English in shelter | dhs | county_em; hiema; hspls; doh; oha; faith_based; other_agencies | — | confirm with partner |
| Interpreters – Chinese | 4 | people with limited English in shelter | dhs | county_em; hiema; hspls; doh; oha; faith_based; other_agencies | — | confirm with partner |
| Interpreters – Hawaiian | 4 | people with limited English in shelter | dhs | county_em; hiema; hspls; doh; oha; faith_based; other_agencies | — | confirm with partner |
| Interpreters – Marshallese | 4 | people with limited English in shelter | dhs | county_em; hiema; hspls; doh; oha; faith_based; other_agencies | — | confirm with partner |

## 6. Decision points

Proposed – supports, does not replace, county plans.

| When | Trigger | Action | Owner (proposed) | Support |
|---|---|---|---|---|
| **BEFORE** | Annual pre-positioning review (no official product)<br>County (sets the date) | Mark coastal shelters in V/A zones for site check; choose inland alternates. | County Emergency Management (each county) | doe, dhs |
| **BEFORE** | Coastal Flood Warning<br>NOAA NWS Honolulu | Alert priority coastal tracts; confirm inland shelters and transport. | County Emergency Management (each county) | dot, dhs |
| **AFTER** | Warning cancelled or all clear<br>County / NWS / PTWC | Recovery sheltering and feeding for displaced households. | American Red Cross | hawaii_voad, salvation_army |

## 7. Priority outreach and languages

**Top 10 census tracts by AFN need (tract-level counts only)**

| # | Tract | At risk | Power-dep. | Disability | Limited English | No vehicle | AFN need |
|---|---|---|---|---|---|---|---|
| 1 | 15001020500 | 5,552 | 31 | 836 | 149 | 1,203 | 2,219 |
| 2 | 15001020600 | 6,802 | 45 | 1,261 | 67 | 387 | 1,760 |
| 3 | 15001020300 | 4,167 | 32 | 670 | 31 | 991 | 1,725 |
| 4 | 15001021300 | 7,384 | 66 | 777 | 176 | 170 | 1,189 |
| 5 | 15001021605 | 4,240 | 30 | 493 | 400 | 250 | 1,174 |
| 6 | 15001021510 | 5,020 | 50 | 575 | 241 | 197 | 1,063 |
| 7 | 15001020400 | 3,398 | 22 | 488 | 80 | 453 | 1,043 |
| 8 | 15001021608 | 4,381 | 35 | 587 | 334 | 61 | 1,017 |
| 9 | 15001021513 | 5,579 | 38 | 568 | 235 | 134 | 975 |
| 10 | 15001021101 | 2,996 | 27 | 550 | 26 | 135 | 738 |

**Languages: people who speak English less than "very well" (Hawaii)**

| Language | People 5+ | 90% MOE |
|---|---|---|
| Ilocano | 3,409 | ±869 |
| Other English-based Creole languages | 2,040 | ±740 |
| Japanese | 1,408 | ±504 |
| Spanish | 1,217 | ±499 |
| Tagalog, Filipino | 924 | ±368 |
| Chinese | 651 | ±326 |
| Hawaiian | 637 | ±264 |
| Marshallese | 538 | ±303 |

Source: ACS PUMS 2019-2023 5-year (validated against DBEDT Data Book 2024, Table 1.44). Pre-approved message templates: none yet; no machine translation without human review.

## 8. Sources and assumptions

### Planning factors

| Factor | Value | Source or team assumption |
|---|---|---|
| Shelter-seeking rate – Hurricane | 0.02 / 0.03 / 0.05 | South Carolina EMD, 2026 Hurricane Response Annex G (planning factors 2%, 3%, 5%) |
| Shelter-seeking rate – Wildfire | 0.02 / 0.05 / 0.112 | high = observed Maui 2023: 1,418 people in shelters on Aug 11, 2023 (Maui Now) / Lahaina CDP population 12,702 (2020 Census) = 0.1116, rounded to 3 decimals = 0.112 (so the calibration event falls inside the band). low and central are team_assumption. |
| Shelter-seeking rate – Tsunami | 0.02 / 0.05 / 0.11 | team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Shelter-seeking rate – Inland flooding | 0.02 / 0.05 / 0.11 | team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Shelter-seeking rate – Coastal flooding | 0.02 / 0.05 / 0.11 | team_assumption: central 5%; low 2% = hurricane planning factor (SCEMD); high 11% = observed Maui 2023 no-notice total-loss rate |
| Footprint – Hurricane | all | Hurricanes affect whole islands (Lala, Lowell 2026). |
| Footprint – Tsunami | tsunami_zone_share | Population share inside the State standard tsunami evacuation zone. Extreme zone is reported separately. |
| Footprint – Wildfire | nri_top_share_by_county (25%) | Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption). |
| Footprint – Inland flooding | nri_top_share_by_county (25%) | Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption). |
| Footprint – Coastal flooding | nri_top_share_by_county (25%) | Counties lead mass care, so each county plans for its own highest-risk tracts (team_assumption). |
| floor sqft 72h | 20 | Florida DEM Sheltering Guidance Aid (2010): emergency evacuation shelter, <=72 h |
| floor sqft 2wk | 40 | Florida DEM (2010): short-term shelter, <=2 weeks |
| water gal per day | 1 | Florida DEM (2010) |
| meals per day | 2 | Florida DEM (2010): two hot meals a day plus snacks |
| persons per toilet short term | 20 | Florida DEM (2010): short-term shelter, <=2 weeks |
| power outlets per power dependent person | 1 | team_assumption |
| persons per toilet 72h | 40 | Florida DEM (2010): emergency evacuation shelter, <=72 h |
| Staffing ratios | client registration 1:165, dormitory management 1:100, meal service 1:110, mental health 1:250, public health medical 1:200, security 1:100, sanitation 1:200, children area 1:12 | One staff member per N shelter residents, minimum 1 when the shelter is open. Florida DEM (2010) staffing metrics; the source marks them as recommendations. |
| dog owning household share | 0.446 | AVMA U.S. pet ownership statistics (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Dog ownership increased by 6% from 38.4% (2016) to 44.6% (2020)'. The first url did not resolve when checked 2026-10-27. |
| cat owning household share | 0.26 | AVMA (2020 data). Verified in 2022 AVMA Pet Ownership and Demographics Sourcebook, p. 3: 'Cat ownership increased slightly from 25% in 2016 to 26% in 2020'. |

### Data sources

| Source | Fetched | URL / file |
|---|---|---|
| CDC/ATSDR Social Vulnerability Index 2022, census tracts | 2026-10-27 | https://onemap.cdc.gov/onemapservices/rest/services/SVI/CDC_ATSDR_Social_Vulnerability_Index_2022_USA/FeatureServer/2 |

| Source | Fetched | URL / file |
|---|---|---|
| FEMA National Risk Index v1.20, census tracts – Hurricane | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Hurricane_Hazard_Type_Risk_Index_ |
| FEMA National Risk Index v1.20, census tracts – Wildfire | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Wildfire_Hazard_Type_Risk_Index_Ra |
| FEMA National Risk Index v1.20, census tracts – Tsunami | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Tsunami_Hazard_Type_Risk_Index_R |
| FEMA National Risk Index v1.20, census tracts – Inland Flooding | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Inland_Flooding_Hazard_Type_Risk_I |
| FEMA National Risk Index v1.20, census tracts – Coastal Flooding | 2026-10-27 | https://services.arcgis.com/XG15cJAlne2vxtgt/arcgis/rest/services/National_Risk_Index_Census_Tract_Coastal_Flooding_Hazard_Type_Risk_ |
| Tsunami Evacuation Zones (Statewide GIS) | 2026-10-27 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/2 |
| Extreme Tsunami Evacuation Zones (Statewide GIS) | 2026-10-27 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/12 |
| HHS emPOWER – electricity-dependent DME, county level | 2026-10-27 | https://services2.arcgis.com/ZQ4jTQn6k7VPXEwO/arcgis/rest/services/HHS_emPOWER_REST_Service_Public/FeatureServer/2 |
| Hawaii Statewide GIS – State DFIRM (FEMA flood hazard areas, 6/15/26) | 2026-10-27 | https://geodata.hawaii.gov/arcgis/rest/services/Hazards/MapServer/6 |

| Source | Fetched | URL / file |
|---|---|---|
| County shelter activations 2026 (county releases and news) | see table | data/sources/county_shelter_activations.csv |
| ACS PUMS 2019-2023 5-year (detailed languages) | 2026-10-27 | https://www2.census.gov/programs-surveys/acs/data/pums/2023/5-Year/csv_phi.zip |
| DBEDT State of Hawaii Data Book 2024, Table 1.44 (validation) | 2026-10-27 | https://files.hawaii.gov/dbedt/economic/databook/2024-individual/01/014424.pdf |
| Partner list: Hawaii DHS, HAIC 2026 ESF-6 challenge statement | 2026-10-15 | HAIC_AI Challenge_Submission_DHS_Disaster_2026.pdf |
| pre season hurricane | | https://www.nhc.noaa.gov/cphcmission.php |
| hurricane watch | | https://www.nhc.noaa.gov/aboutgloss.shtml |
| hurricane warning | | https://www.nhc.noaa.gov/aboutgloss.shtml |
| tsunami warning | | https://www.tsunami.gov/?page=message_definitions |
| flash flood warning | | https://www.weather.gov/safety/flood-watch-warning |
| red flag warning | | https://www.weather.gov/safety/wildfire-ww |
| coastal flood warning | | https://www.weather.gov/okx/marinedef |

Privacy: aggregate census-tract and county data only. No individual records. Public facility names and addresses only.